

Impact of Health-Related Variables on Insurance Premiums

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Research Topic and Background

01



01

Research Topic and Background

Background

How do health-related variables (BMI, smoking status) affect insurance premiums?

- Risk imposes economic costs on individuals and inefficiencies on society
- From the individual's perspective, **insurance** serves as a hedge against such risk.
- From the insurer's perspective, it is crucial to set appropriate premiums for consumers with varying risk levels.
- Among **health-related variables**, *smoking status* and *BMI* are expected to be most relevant to premiums.
- This study aims to examine how smoking and BMI significantly affect insurance premiums and to determine which model best explains premiums when including other individual-specific characteristics.

Variable Transformation and Definition

02



02

Variable Transformation and Definition

Interaction vs Group Dummy Variable

Based on smoking status and obesity, the entire dataset can be grouped into four categories.

	BM <30	BM ≥30
Smoke X	Group1	Group2
Smoke 0	Group3	Group4

When considering the interaction between smoking and BMI, the variables affecting the four groups are as follows.

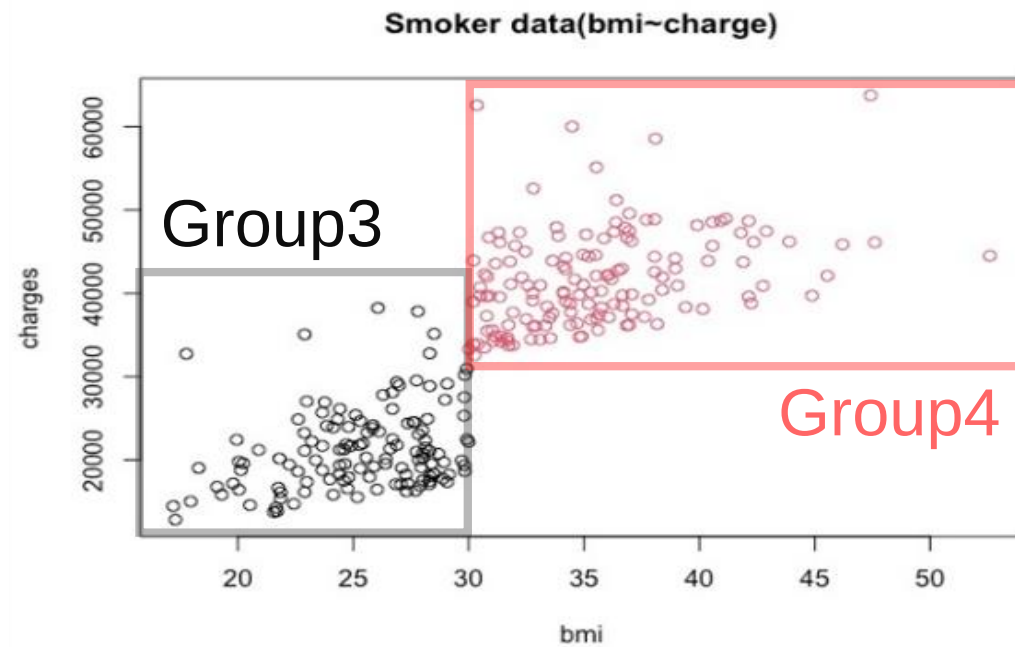
	BM <30	BM ≥30
Smoke X	bmi	bmi
Smoke 0	smoke*bmi + smoke + bmi	smoke*bmi + smoke + bmi

02

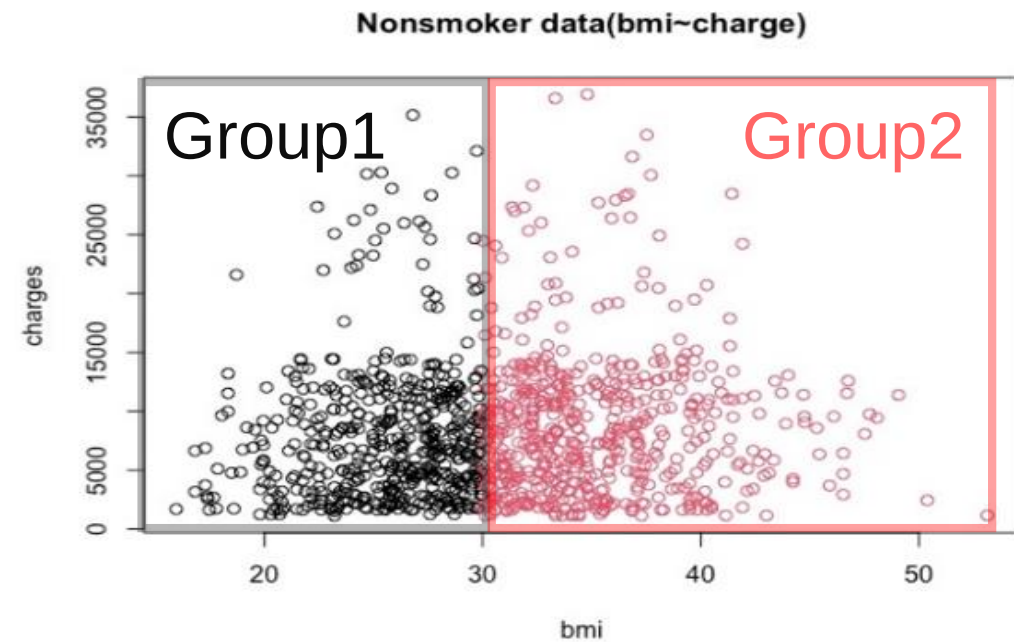
Variable Transformation and Definition

Interaction vs Group Dummy Variable

smoker



Nonsmoker



◆ Smokers show a distinct difference at BMI 30, while nonsmokers do not.

Using an interaction term may not adequately explain Groups 1 and 2.

02

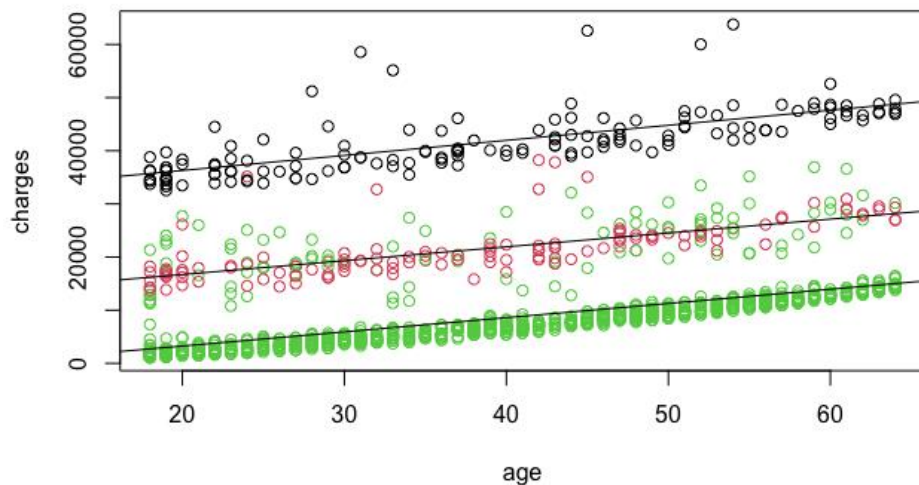
Variable Transformation and Definition

Interaction vs Group Dummy Variable

To address this, groups showing no clear difference were merged, creating three total groups.

	BM <30	BM ≥30
Smoke X	Group3	
Smoke 0	Group2	Group1

A scatter plot of the dependent variable (*charges*) and *age* showed significant differences by group.



>>

- Hence, **dummy variables** were introduced to allow different intercepts for each group.
- Hence, dummy variables were introduced to allow different intercepts for each group.

02

Variable Transformation and Definition

BMI Centering

A regression model including the interaction term $bmi \times smoke$ initially

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2223.454	865.611	-2.569	0.01032	*
age	263.620	9.516	27.703	< 2e-16	***
sexmale	-500.146	266.518	-1.877	0.06079	.
bmi	23.533	25.601	0.919	0.35814	
smoke	-20415.611	1648.277	-12.386	< 2e-16	***
children	516.403	110.179	4.687	3.06e-06	***
regionnorthwest	-585.478	380.859	-1.537	0.12447	
regionsoutheast	-1210.131	382.750	-3.162	0.00160	**
regionsouthwest	-1231.108	382.218	-3.221	0.00131	**
bmi:smoke	1443.096	52.647	27.411	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- counterintuitive negative coefficient for *smoke*
- Because the interaction effect was very large, the interpretation became difficult.
- To mitigate this, *BMI centering* was performed.

centering

```
> summary(data$bmi)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
15.96	26.30	30.40	30.66	34.69	53.13

- Median: 30.4, Mean: 30.66
- Each BMI value was centered by subtracting the median.

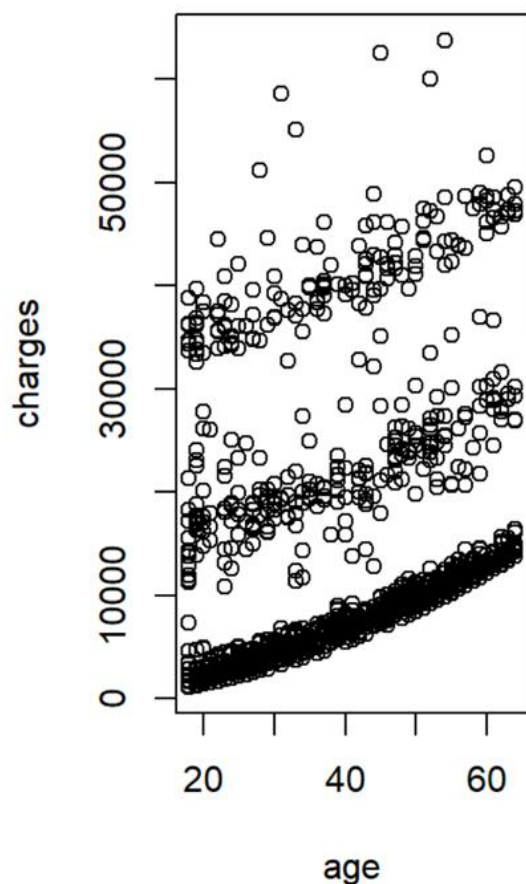
Since the obesity threshold (BMI 30) is close to the median, negative centered BMI values indicate *normal weight*, while positive values indicate *obesity*.

02

Variable Transformation and Definition

Polynomial Age Terms (age^2 , age^3)

charges~age



- The relationship between *charges* and *age* appeared quadratic or cubic. Including the age^3 term did not change explanatory power and made *age* itself insignificant.
- The model including **only the age^2 term showed statistical significance**, and an F-test confirmed that age^2 was significant relative to a model with only *age*.

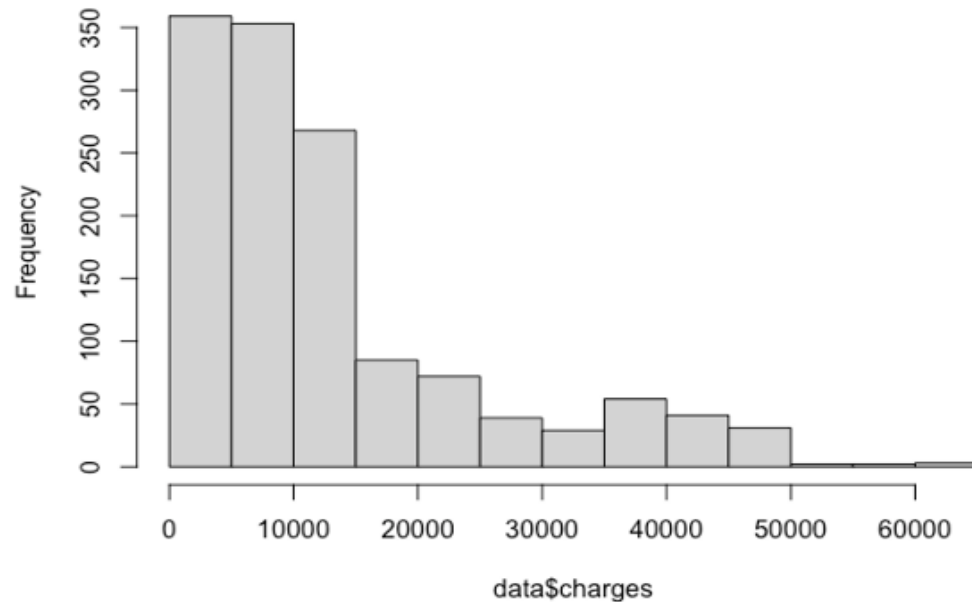
02

Variable Transformation and Definition

Log Transformation of Charges

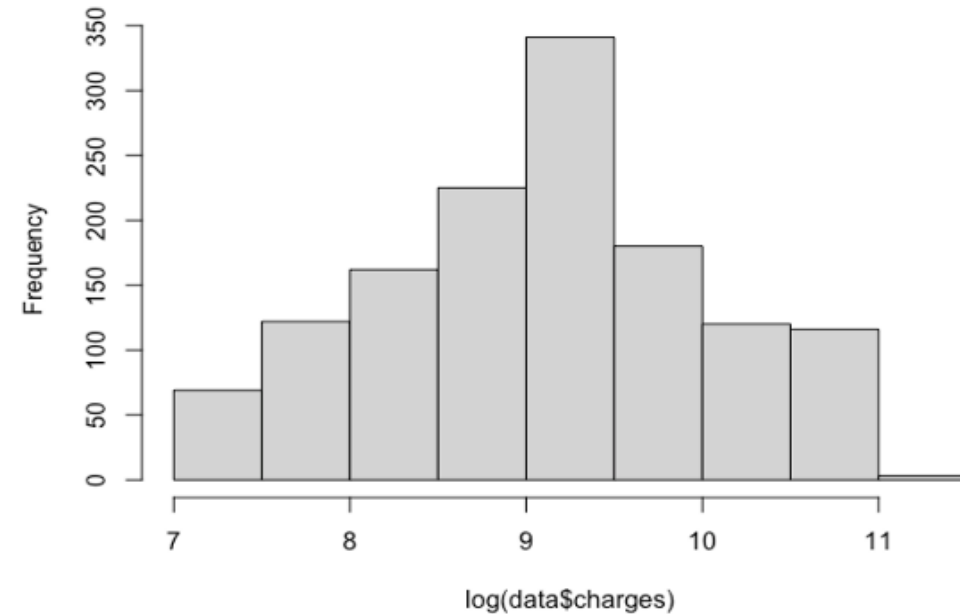
charges

Histogram of Charges



log(charges)

Histogram of log(Charges)



- ◆ The histogram of *charges* showed a right-skewed distribution.
To address potential violation of normality in residuals, the dependent variable **was log-transformed**.
- ◆ After transformation, the distribution **resembled normality**.

Analytical Models

03



03

Analytical models

Nine models were constructed with different variable combinations, including:

1	All Variable Regression		
2	Group Dummy Variable		
3	Smoke * Bmi Interaction		
4	Smoke * Bmi Interaction	Age^2	
5	Group Dummy Variable	Age^2	
6	log(Charges)	Smoke * Bmi Interaction	
7	log(Charges)	Group Dummy Variable	
8	log(Charges)	Smoke * Bmi Interaction	Age^2
9	log(Charges)	Group Dummy Variable	Age^2

Regressor	Y = charges					Y = log(charges)			
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
age	256.9 *** (11.9)	263.6 *** (8.758)	263.6 *** (9.5)	-33.2 *** (64.6)	-12.9 (59.4)	0.03 *** (<0.001)	0.03 *** (<0.001)	0.05 *** (0.005)	0.05 *** (0.005)
ag2^2				3.7 (0.8)	3.4 *** (0.7)			-0.0002 ** (<0.001)	-0.0002 *** (<0.001)
bmi_med	339.2 *** (28.6)	59.8 ** (22.6)	23.5 (25.6)	20.1 (25.4)	57.1 * (22.4)	0.003 (0.002)	0.004 * (0.002)	0.003 (0.002)	0.004 * (0.002)
smoke	23848.5 *** (413.1)		23454.5 *** (330.6)	23465.7 *** (328.1)		1.54 *** (0.02)		1.54 *** (0.02)	
bmi_m*smoke			1443.1 *** (52.6)	1441.1 *** (52.2)		0.04 *** (0.004)		0.04 *** (0.004)	
sexmale	-131.3 (332.9)	-470.8 (245.2)	-500.1 (266.5)	-506.4 (264.5)	-476.3 (243.2)	-0.08 *** (0.02)	-0.086 *** (0.02)	-0.08 *** (0.02)	-0.085 *** (0.02)
children	475.5 *** (137.8)	508.5 *** (101.4)	516.4 *** (110.1)	675.1 *** (114.5)	656.3 *** (105.4)	0.1 *** (0.01)	0.1 *** (0.01)	0.09 *** (0.01)	0.09 *** (0.01)
regionnorthwest	-353.0 (476.3)	-274.8 (350.4)	-585.4 (380.8)	-599.3 (377.9)	-288.2 (347.7)	-0.07 * (0.03)	-0.06 (0.03)	-0.07 * (0.03)	-0.06 (0.03)
regionsoutheast	-1035.0 * (478.7)	-784.1 * (352.3)	-1210.1 ** (382.7)	-1206.5 ** (379.8)	-781.6 * (349.5)	-0.16 *** (0.03)	-0.14 *** (0.03)	-0.01 *** (0.03)	-0.14 *** (0.03)
regionsouthwest	-960.0 * (477.9)	-1154.5 ** (351.7)	-1231.1 ** (382.2)	-1228.3 ** (379.3)	-1151.8 *** (348.9)	-0.13 *** (0.03)	-0.13 *** (0.03)	-0.01 *** (0.03)	-0.13 *** (0.03)
g1		33177 *** (411.9)			33164.2 *** (408.7)		1.85 *** (0.04)		1.85 *** (0.04)
g2		13577.5 *** (431.4)			13612.5 *** (428.1)		1.22 *** (0.04)		1.22 *** (0.04)
(intercept)	-1627.1 ** (624.8)	-1728.5 *** (459.8)	-1508.1 ** (499.5)	3467.7 ** (1180.3)	2908.2 ** (1086.4)	7.44 *** (0.04)	7.43 *** (0.04)	7.12 *** (0.1)	7.1 *** (0.1)
SER	6062	4461	4846	4809	4426	0.4293	0.4242	0.4277	0.4225
Adjusted R^2	0.7494	0.8643	0.8398	0.8423	0.8664	0.7821	0.7871	0.7837	0.7889

Variable Selection and Model Significance Testing

04



04

Variable Selection and Model Significance test

Process Overview

■ Process of Variable Selection

- Variables were sequentially removed, and metrics such as Adjusted R^2 , Cp, AIC, and BIC were computed.

Since AIC penalizes variable count less heavily than BIC and the dataset has few variables, AIC was used for model selection.

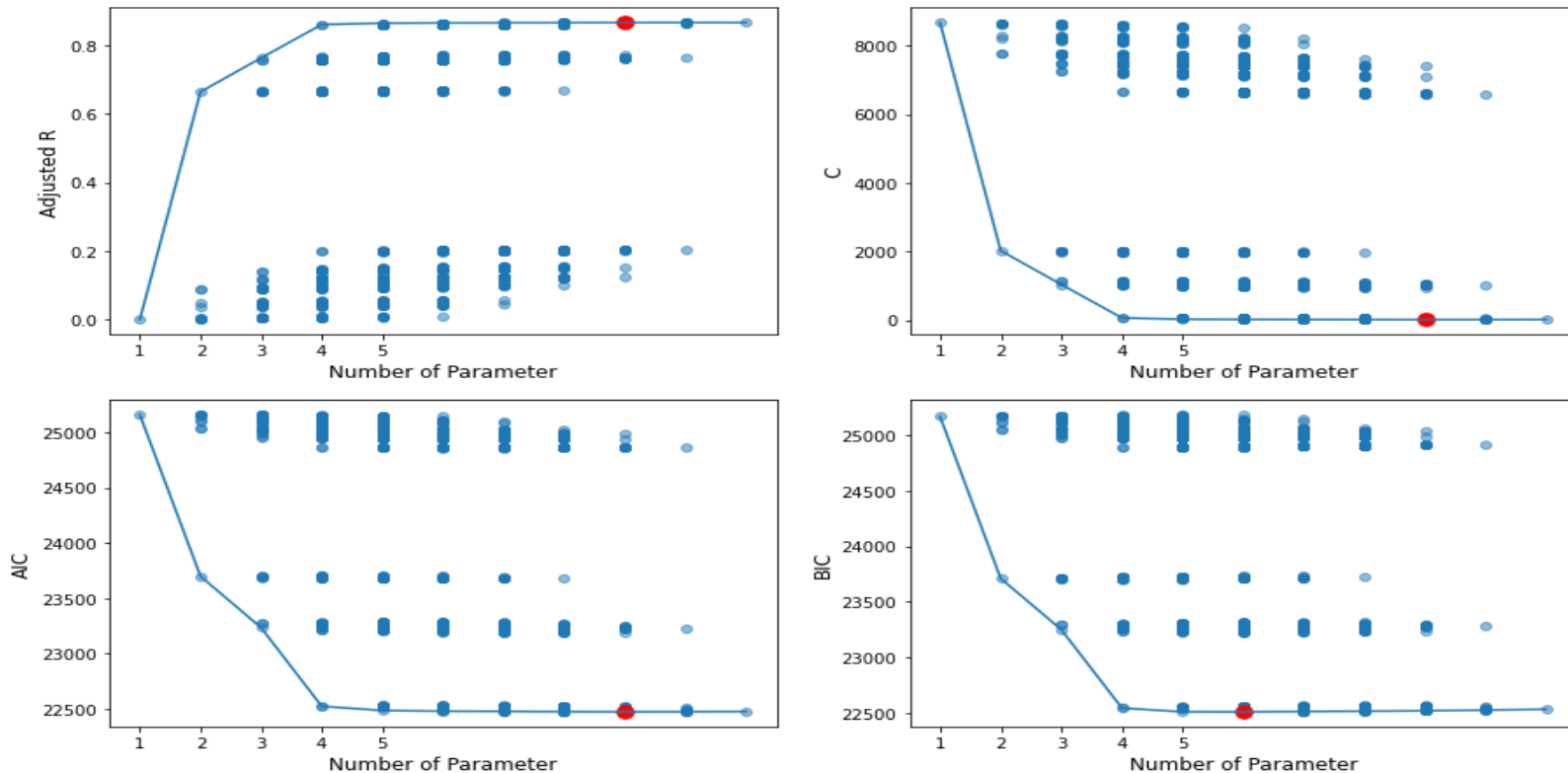
■ Model Significance test

- The model with the optimal AIC was defined as the *Reduced Model*, and an F-test was conducted against the *Full Model* to test significance.

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Variable Selection and Model Significance test

Model 5



Variables

- g1
- g2
- age_2
- sex_male
- bmi_med
- children
- region_southeast
- region_southwest

→ F-test result: No significant difference

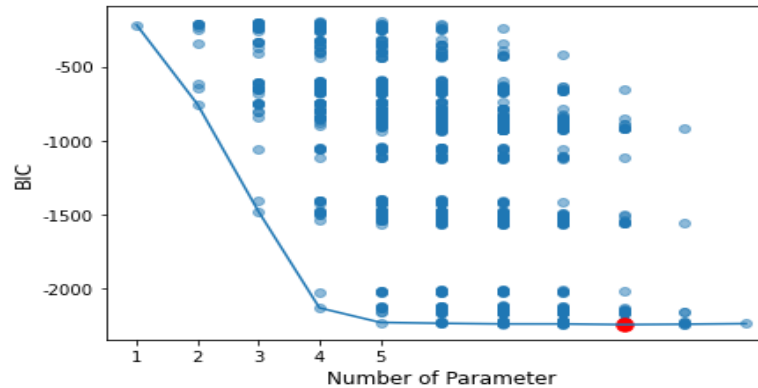
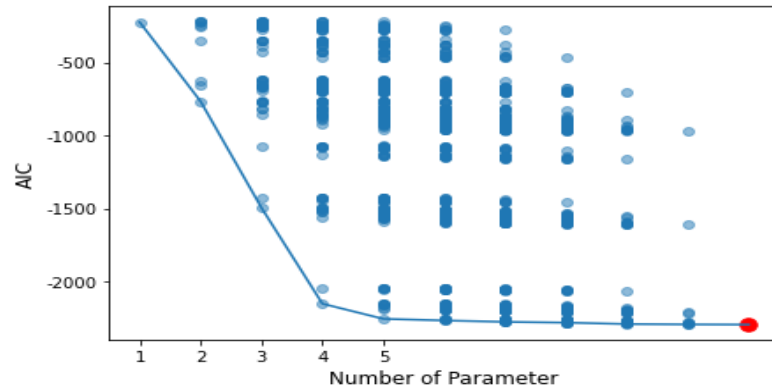
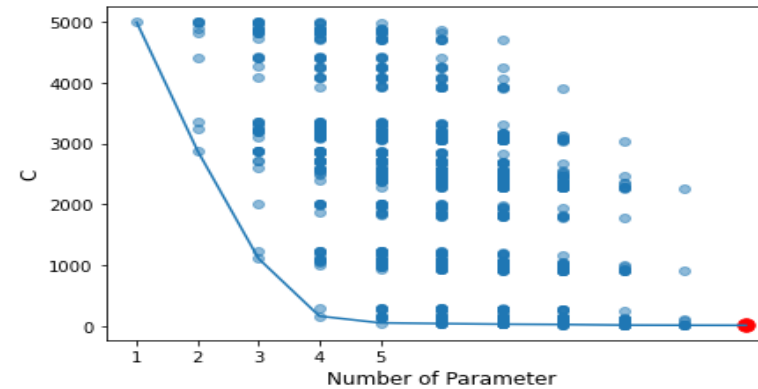
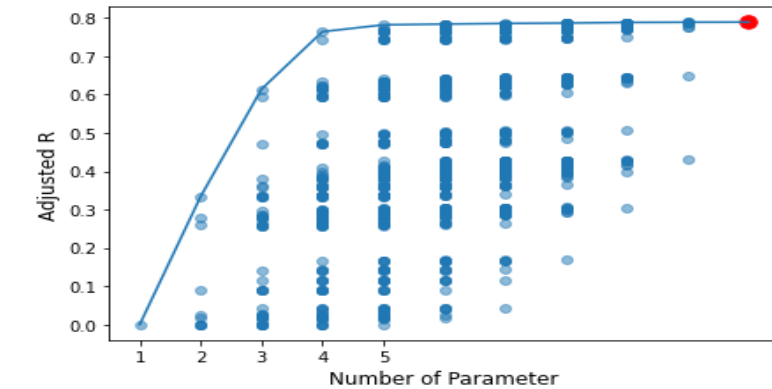
between Full and Reduced Models.

	Variables	Number_of_parameter	Ad_R	Cp	AIC	BIC
991	g1, g2, age_2, sex_male, bmi_med, children, re...	9	0.866554	7.731577	22473.345682	22520.136063
418	g1, g2, age_2, children, region_southwest	6	0.865636	13.884171	22479.536556	22510.730143
1023	g1, g2, age, age_2, sex_male, bmi_med, childre...	11	0.866426	11.000000	22476.608244	22533.796487

04

Variable Selection and Model Significance test

Model 9



Variables

- g1
- g2
- age
- age_2
- sex_male
- bmi_med
- children
- region_northwest
- region_southeast
- region_southwest

	Variables	Number_of_parameter	Ad_R	Cp	AIC	BIC
1023	g1, g2, age, age_2, sex_male, bmi_med, childre...	11	0.788873	11.000000	-2294.537505	-2237.349262
976	g1, g2, age, age_2, sex_male, children, region...	9	0.787950	14.814225	-2290.681612	-2243.891231

Regression Diagnostics and Conclusion

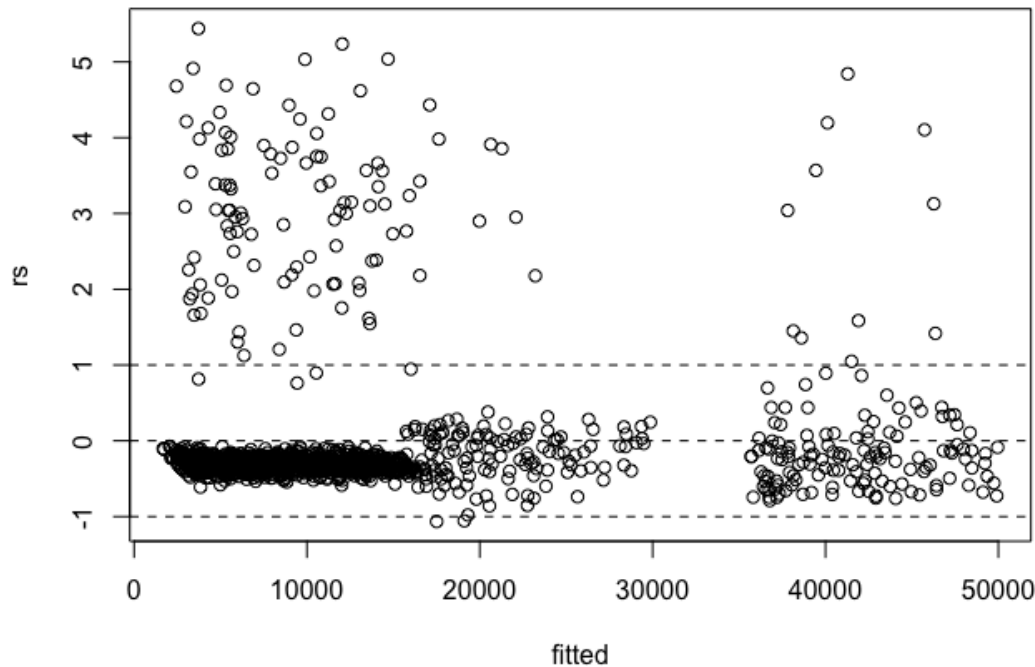
05



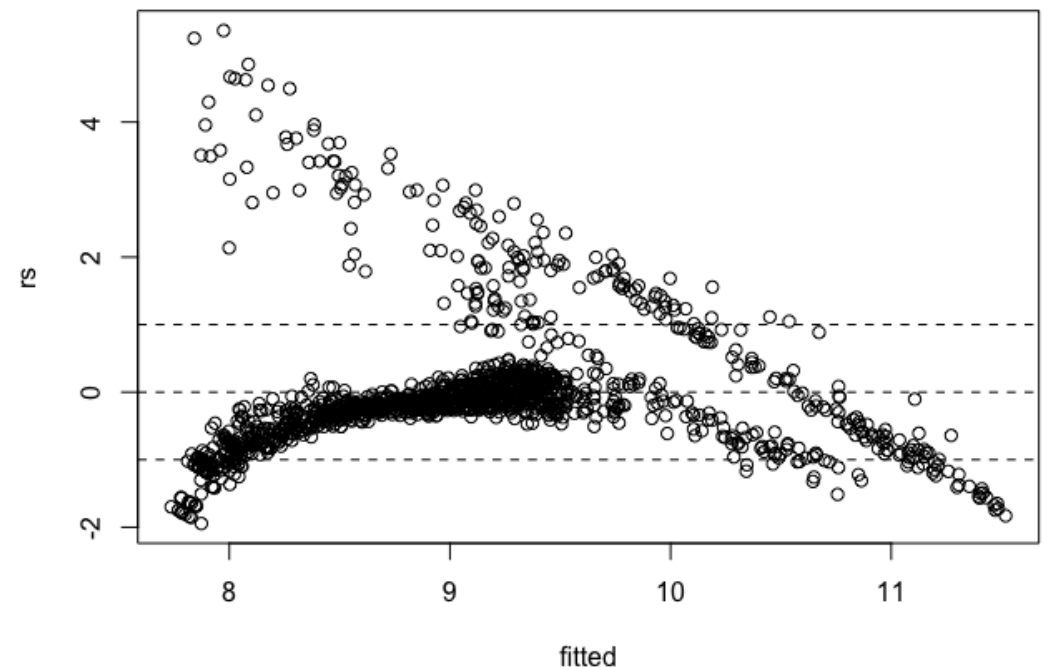
05

Regression Diagnostics and Conclusion

Residual Analysis – Homoscedasticity

Model 5**Standardized Residual Plot**

- No distinct trend in residuals versus fitted values

Model 9**Standardized Residual Plot**

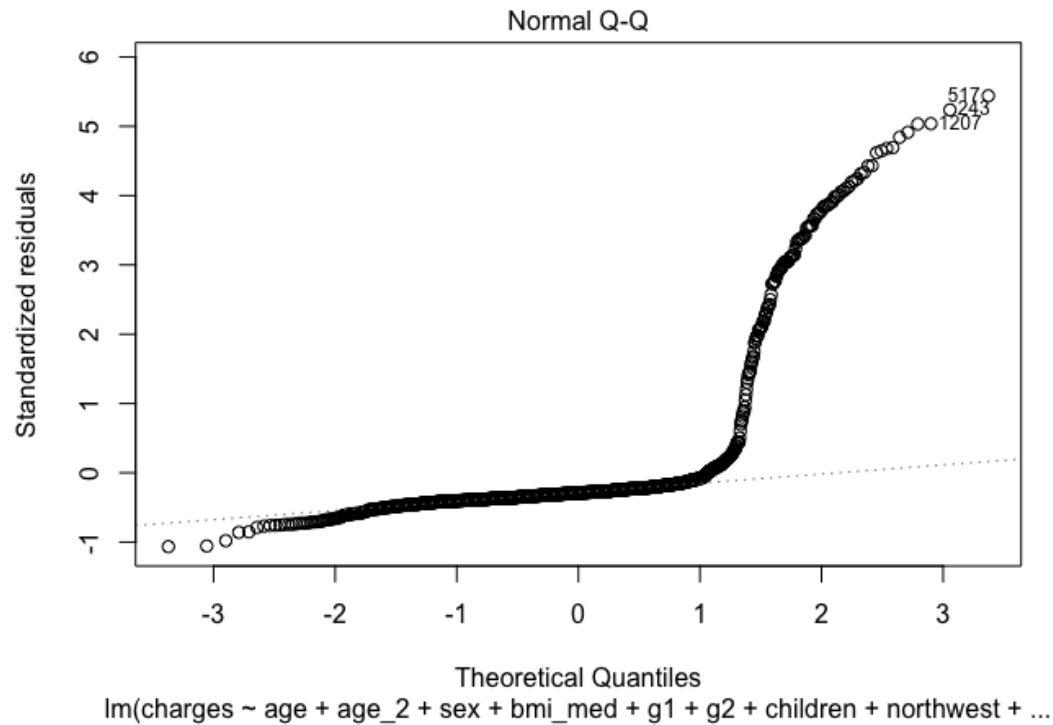
- Residual variance decreased as fitted value increased

05

Regression Diagnostics and Conclusion

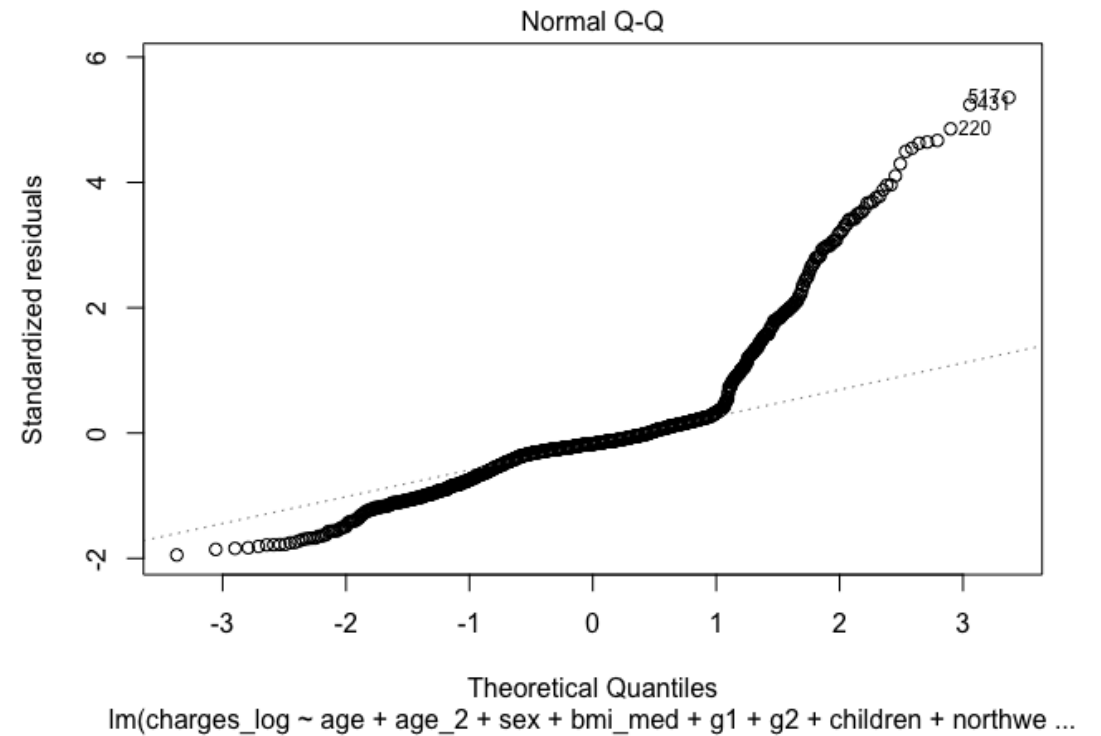
Residual Analysis – Normality

Model 5



- Normality assumption violated

Model 9



- Normality assumption violated

05

Regression Diagnostics and Conclusion

Multicollinearity (VIF)

Model 5

```
> vif(lm5)
```

age_2	bmi_med	sex	g1	g2	children	southeast	southwest
1.016333	1.280502	1.010560	1.095187	1.097457	1.001592	1.244835	1.147982

Model 9

```
> vif(lm9)
```

age	age_2	sex	bmi_med	g1	g2	children	northwest
47.629882	47.582251	1.010644	1.280637	1.095878	1.097962	1.102096	1.518992
southeast	southwest						
1.652978	1.529831						

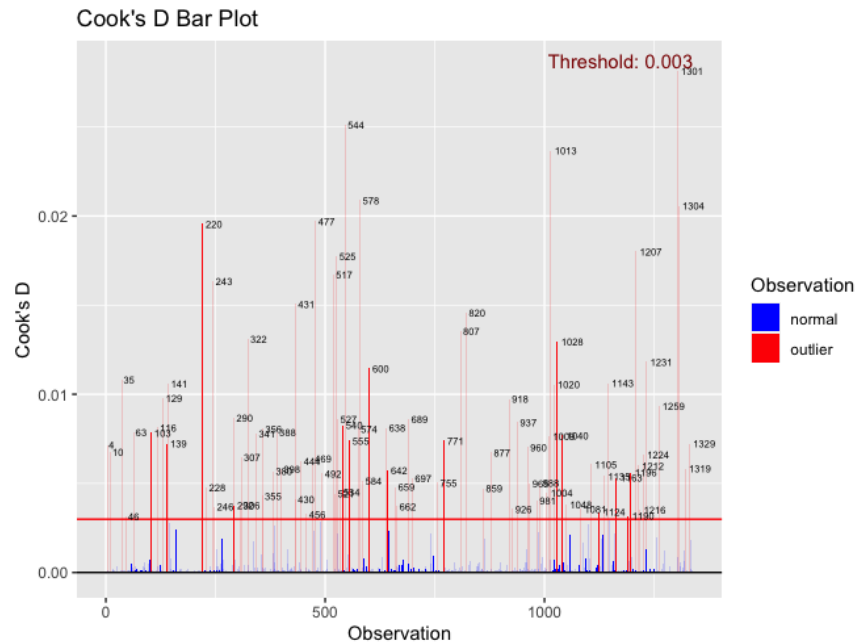
◆ No multicollinearity issue in either model.

◆ Although *age* and *age*² had high VIF in Model 9, this was not problematic since they are polynomial terms.

Regression Diagnostics and Conclusion

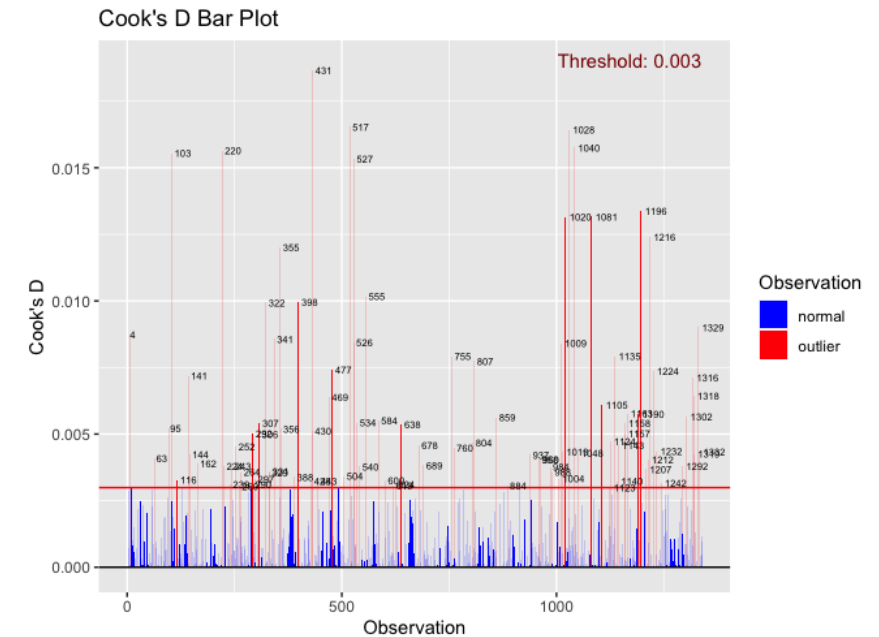
Outliers (Cook's Distance)

Model 5



- Many extreme values detected

Model 9



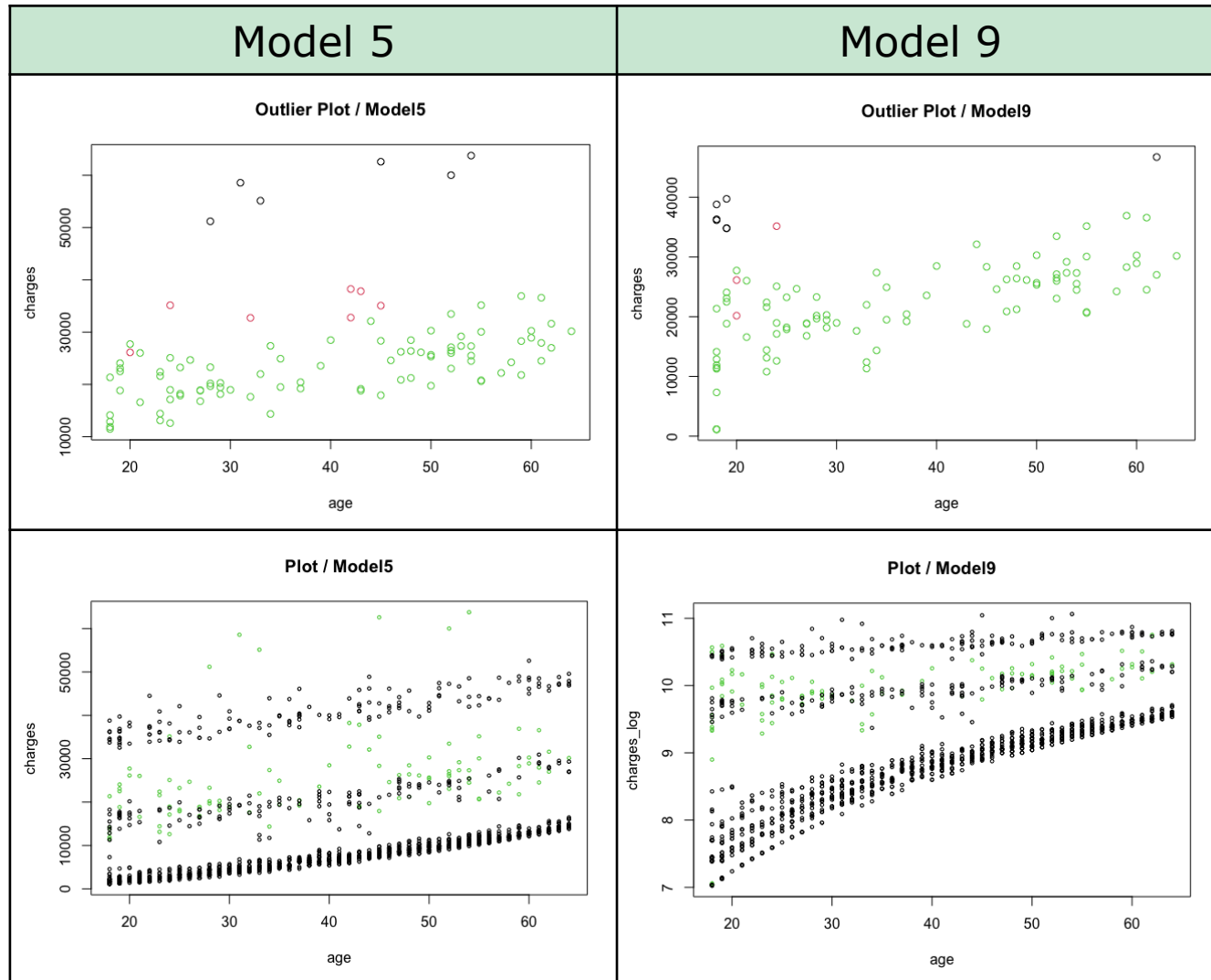
- Many extreme values detected

- ◆ Data points in Group 3 with *high charges* correspond to outliers in *charges–age* scatter plots.

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Regression Diagnostics and Conclusion

Outliers (Cook's Distance)



- Plotting top 100 outliers confirmed **most belong to Group 3 (green points)**.
- While removing them slightly increased R^2 , **removal was not justified; thus, all were retained**.

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Regression Diagnostics and Conclusion

Final Model Selection and Interpretation

Final Model Selection and Interpretation

Both models (5 and 9) showed similar diagnostic and outlier patterns.

- Explanatory power: **Model 5 (0.8666)** > Model 9 (0.7889)
- Model 9's variables were more individually significant.
- Model 9 explains the *rate of premium increase*, while Model 5 explains the *premium level*.

Given the research goal—to identify a model that explains individual risk—**Model 5 was selected as final.**

If the study aimed to analyze premium growth (flow) rather than premium level (stock), Model 9 might have been chosen. However, due to the **limited number of variables and their static nature**, Model 9's interpretation was less appropriate.

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Regression Diagnostics and Conclusion

Interpretation of Final Model

```
> lm<- lm(charges~age_2+bmi_med+sex+g1+g2+children+southeast+southwest)
> summary(lm)
```

Call:

```
lm(formula = charges ~ age_2 + bmi_med + sex + g1 + g2 + children +
    southeast + southwest, data = data)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-4491.0 -1637.2 -1290.3  -846.8 24038.4
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2540.7657	313.1874	8.113	1.12e-15	***
age_2	3.3276	0.1083	30.714	< 2e-16	***
bmi_med	57.0111	22.4503	2.539	0.011216	*
sexmale	-475.1358	243.1669	-1.954	0.050917	.
g1	33172.8198	408.4006	81.226	< 2e-16	***
g2	13615.3777	427.7739	31.828	< 2e-16	***
children	646.9562	100.4414	6.441	1.65e-10	***
southeast	-637.6871	303.2156	-2.103	0.035646	*
southwest	-1007.7633	302.1669	-3.335	0.000876	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4424 on 1329 degrees of freedom

Multiple R-squared: 0.8674, Adjusted R-squared: 0.8666

F-statistic: 1086 on 8 and 1329 DF, p-value: < 2.2e-16

- Regression results show that individuals who are both obese and smokers exhibit higher personal risk.
- According to the “*Korean Journal of Nursing Science*”, smokers have twice the risk of coronary artery disease as nonsmokers, and those who are both obese and smokers have seven times the risk of hyperlipidemia compared to normal-weight nonsmokers.
- These findings are consistent with the study’s results.

Limitations and Strengths

06



06

Limitations and Strengths

Limitations

Omitted Variable Problem (1)

- In the U.S., private health insurance is often tied to employment, meaning income or job-related factors may affect premiums.

However, the **dataset lacks such economic variables (income, employment)**, preventing estimation of economic effects.

Omitted Variable Problem (2)

- Some data show implausible child counts (e.g., 5 children with parent age 18), but this could occur due to adoption practices.

Yet, without marital status variables, **omitted variable bias remains.**

06

Limitations and Strengths

Limitations

Difficulty Using Life Tables

- Insurance companies typically use life tables based on death probabilities (a flow variable).
Due to missing key features (race, employment status, etc.), this dataset cannot fully reflect such actuarial structures.

Dataset Limitations

- Income and premium distributions vary across U.S. states, but this dataset classifies only into four broad regions (NE, NW, SE, SW), limiting granularity.

06

Limitations and Strengths

Limitations

Differences between U.S. and Korean Insurance Markets

- While Korean insurance emphasizes public welfare, the U.S. market is largely private and risk-based.

Thus, the model developed here may explain U.S. insurer risk well but cannot be directly applied to the Korean context.

06

Limitations and Strengths

Strengths

Use of Diverse Variable Transformations

- Interaction terms, group dummy variables, polynomial terms, and log transformations were applied to construct models from multiple perspectives.

Multiple Indices and Visualizations for Variable Selection

- Adjusted R^2 , AIC, BIC, and C_p metrics were used to enhance reliability.

Practical Relevance:

- Model interpretations were consistent with findings from the *Korean Journal of Nursing Science*, supporting real-world applicability.

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References

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<Vrieze, S. I. (2012), "Model selection and psychological theory: a discussion of the differences between the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC)",>

- **Health / BMI Criteria:**

Centers for Disease Control and Prevention (CDC). (2011). *Adult BMI Classification and Health Implications*. U.S. Department of Health & Human Services.

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World Health Organization (WHO). (2012). *Global Report on Trends in Tobacco Use and Health Outcomes*.

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Rosen, S., & Thaler, R. H. (2001). *Risk and Uncertainty in Insurance Decision-Making*. *Journal of Economic Perspectives*, 15(4), 23–40.

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Korean Journal of Nursing Science. (n.d.). *Comparative Risk of Cardiovascular and Lipid Disorders in Smokers and Obese Individuals*.